Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2024/2025 (Unit 6)

Stephen D. Wolthusen
stephen.wolthusen@rhul.ac.uk

Department of Information Security and Communication Technology
Norwegian University of Science and Technology, Norway

and

Department of Information Security
Royal Holloway, University of London, United Kingdom

Unit 6— Stuxnet Attribution Issues

Contents

Introduction to Olympic Games

Attribution Issues

Attribution

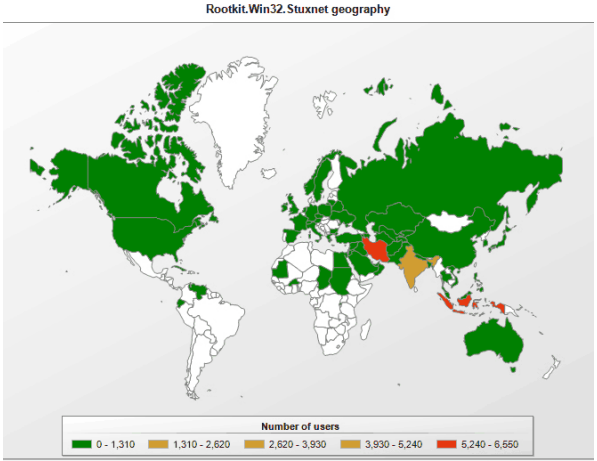
Stuxnet Attack Mechanisms

Part II

Attribution Issues

Attributing Stuxnet

Stuxnet Geographical Distribution Snapshot

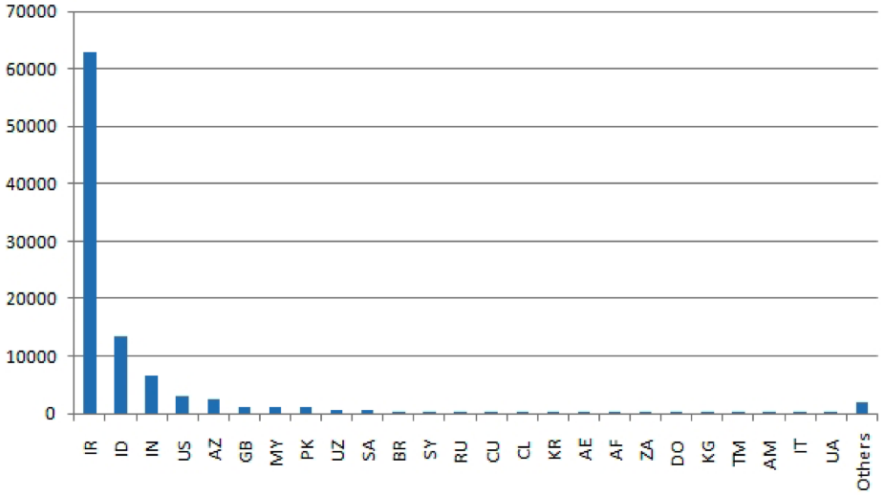


Malware analysis: *Backscatter* traffic for Stuxnet (September 2010)

Source: University of Maryland – Baltimore County (US)

Attributing Stuxnet

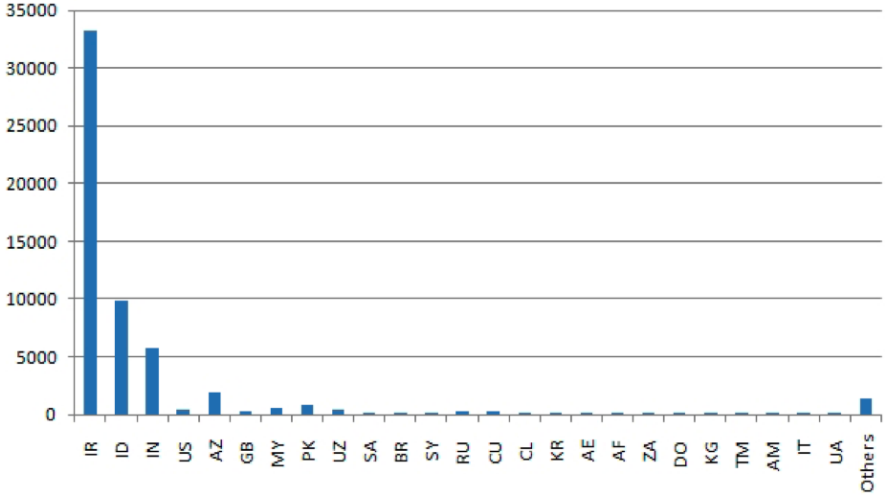
Stuxnet Geographical Distribution Statistics



Number of infected hosts by top-level domain (September 2010)

Attributing Stuxnet

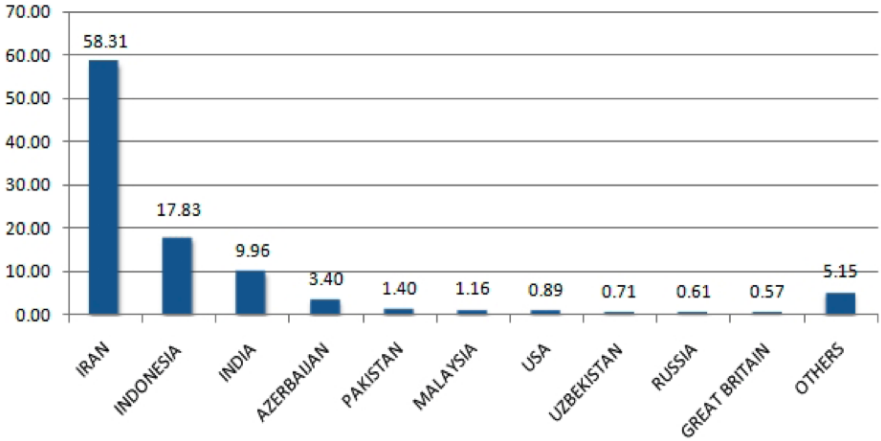
Stuxnet Geographical Distribution Statistics



Infected organisations by TLD (September 2010)

Attributing Stuxnet

Stuxnet Geographical Distribution Statistics

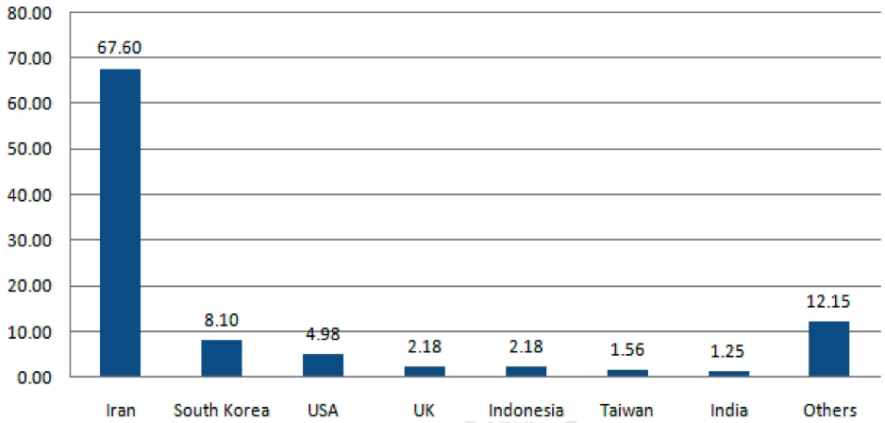


Geographical distributions according to WHOIS data (September 2010)

Source: Symantec

Attributing Stuxnet

Stuxnet Geographical Distribution Statistics



Percentage of Stuxnet-infected hosts having Siemens SS7 software installed (September 2010)

Source: Symantec

Initial Targets

Initial infection in June 2009 affecting *Foolad Technic Engineering Co.*, an Iranian company in Isfahan dealing with industrial automation

Source: Kaspersky Labs

Initial Targets

Second infection waves in April 2010 affected *Foolad Technic Engineering Co.* and *Behpajooh Co. Elec. & Comp. Engineering*, also dealing with industrial automation; subsequent attacks also affected companies in Iran on the U.N. sanctions list

Source: Kaspersky Labs

Initial Targets

The most likely origin of the worldwide propagation of Stuxnet is assumed to be *Behpajooh Co.*

- Other targets of interest included *Neda Industrial Group*, which is believed to be the developer of the Iranian variant (IR-1) of an original (stolen by A.Q. Khan in 2004) Urenco design

```
J:\Work\Stuxnet\infectlog\logs\neda
==== samples/1875 === found .stub 00002600 [00093000]
 samples/1875 2009.07.07 win 5.2 APPLSERVER NEDA
 samples/1875 2009.07.07 win 5.2 APPLSERVER NEDA
 samples/1875 2009.07.07 win 5.2 WINROUTE NEDA
 samples/1875 2009.07.13 win 5.2 WINROUTE NEDA
 samples/1875 2009.07.13 win 5.2 CRM-SRV NEDA
 samples/1875 2009.07.30 win 5.2 CRM-SRV NEDA
```

Source: Kaspersky Labs

A High-Level Overview of Stuxnet

Deployment Stage

The Windows component of Stuxnet consists of a large DLL with 32 exports and 15 resources. These provide

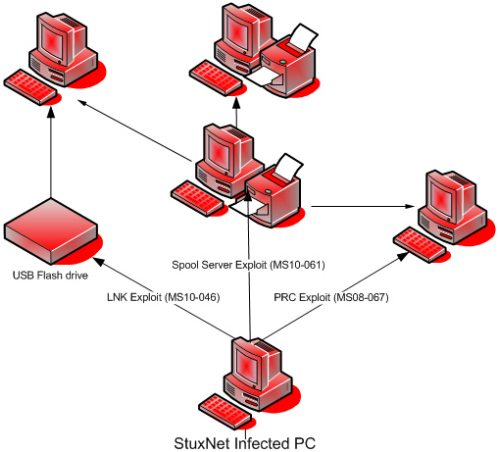
- External (remote) command and control services
- Actual payload infecting **Siemens Simatic Step 7** projects
- Propagation mechanisms

The attack vectors exploit print spoolers remotely, local LNK files (typically via USB pen drives), and via network (Microsoft SMB protocol) shares

- Operating system variants from Windows 2000 onwards (including XP, Vista, 7 and server variants 2008 and 2008R2) were catered for explicitly — obviously this was prior to the release of Windows 10 (and 8)

A High-Level Overview of Stuxnet

Attack Vectors



Multiple overlapping attack vectors for air-gapped and networked systems

Source: Kaspersky Labs

Summary

This section provided material on

- Attribution By Network Analysis
- Tracing Code